Kazi Ababil Azam

Mail Website LinkedIn Github Google Scholar

Work Experience

Software Engineer I, Synesis IT PLC *Software Development, React, Go, TypeScript* Jun 2024 to present
Dhaka, Bangladesh

Contributed to the development of "Convay", a video conferencing and collaborative web application. Notable codebases include the meeting panel, desktop application (Convay Desktop), implementation of in-meeting annotation over shared screen, and offline file sync for Convay Drive.

Project Intern, ERA InfoTech Limited *Natural Language Processing, Data Analysis, Python* May 2023 to Jun 2023
Dhaka, Bangladesh

Worked on a month-long internship project focused on analysis of real-life business data applying natural language processing techniques.

Education

Bachelor of Science (B.Sc.) in Computer Science & Engineering, Apr 2019 to Jul 2024
Bangladesh University of Engineering & Technology (BUET) *CGPA: 3.67/4.00* Transcript

Notable Courses: Machine Learning, Microprocessor, Microcontroller & Embedded Systems, Computer Networking, Software Engineering, Computer Architecture, Operating System, Simulation & Modelling, Computer Security, Artificial Intelligence

Research Experience

Towards Intelligent Traffic Signaling in Dhaka City Based on Vehicle Detection and Congestion Optimization Jun 2023 to Jun 2024
NSysS 2024, 2023
arXiv

Undergraduate thesis under the supervision of Dr. A. B. M. Alim Al Islam.
Posters presented at NSysS 2023, NSysS 2024. Manuscript submitted to Q1 journal, archived.

Devised a system architecture for traffic signal scheduling that incorporates successive implementations of vehicle detection and multi-objective optimization in a Raspberry Pi, specifically focused on the non-lane-based, heterogeneous traffic of Dhaka.

Implemented the proposed system architecture in a cost-effective manner and tested in a simulated environment and at a five-road intersection, validating its feasibility.

Prepared the NHT-1071 (Non-lane-based and Heterogeneous Traffic) dataset, consisting of images of Dhaka City traffic which involves the procurement of the images, the labeling of the vehicles, and benchmarking against state-of-the-art models. Applied in the context of traffic signaling for heterogeneous traffic observed in developing countries.

Investigating Data Privacy in AI Chatbots May 2025 to present

Under the supervision of Dr. Dipto Das and Dr. Imtiaz Karim.
Manuscript in preparation for submission to a conference.

Collection and thematic analysis of anonymous user reviews in online communities to detect and analyze privacy concerns relevant to romantic and emotional AI chatbots.

Pedestrian Trajectory Categorization and Analysis in AV Simulation Jun 2024 to present

Under the supervision of Dr. Golam Md Muktadir, Dr. A. B. M. Alim Al Islam, and Dr. Fahim Hasan Khan.

Analysis of real-life pedestrian trajectories from datasets in order to formulate metrics for generative trajectory models in autonomous vehicle simulations.

Categorizing trajectories found in public pedestrian trajectory datasets based on human annotation and shape perception of their visual structures through online survey.

Evaluation of Reasoning in Large Language Models Jun 2025 to Aug 2025

Under the supervision of Dr. Golam Md Muktadir

Analysis of thinking capability in large language models (LLMs) and testing the reasoning LLMs on how they solve formulated problems based on absurd world models.

Academic Projects

Contradictory, My Dear Watson: Extended *ML, Natural Language Inference* Github

A project extended from a Kaggle competition to detect contradictions in multilingual text data, including Bangla text, through rigorous preprocessing and fine-tuning.

Implementation of a Congestion Control Algorithm for TCP Flows in ns-2 *ns-2* Github

Implementation of a research paper on congestion control in ns-2 codebase, specifically in TCP to achieve improved throughput in wired and wireless networks.

SyncInc *Django, React, MUI, PostgreSQL* Github

A web-based project and task delegation management software designed for organizations. Stack includes PostgreSQL as the database, Django for the backend, ReactJS with Material UI for the frontend, Django REST framework for API development, and Django Channels for real-time notifications.

Laser Security System With Arduino *Arduino, Cyber Physical Systems* Github | YouTube

A project to control a laser security system with microcontrollers, where the ATMega32 unit manages the laser and the Arduino unit controls the alarm system and overall coordination.

Ray Tracing and Illumination Using OpenGL *C++, OpenGL* Github

Implementation of an image generation pipeline using ray tracing and illumination techniques to create realistic images of geometric shapes (sphere, pyramid, cube) and a 2D plane using OpenGL.

Implementation of a Congestion Control Algorithm for TCP Flows in ns-2 *ns-2* Github

Implementation of a research paper on congestion control in ns-2 codebase, specifically in TCP to achieve improved throughput in wired and wireless networks.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, x86 Assembly, MySQL, Bison, Flex, Bash, Go, TypeScript
Frameworks: React, Bootstrap, Django, Express.js, Next.js, Electron.js, Wails, FastAPI, PRAW
Technologies: LaTeX, Docker, Oracle DBMS, PostgreSQL, Git, Wireshark, Postman, OpenGL, ns-2
Libraries: OpenCV, PyAV, PyTorch, Keras, Pandas, Scikit-learn, SciPy, NLTK, Matplotlib, Seaborn, p5.js

Competitions

Champion Student Poster, 11th NSysS 2024 *Poster Presentation* Poster | Certificate
5th Place, IEEE CS, BUET presents GameJam 2023 *p5.js* Github

Leadership

Director of Events, BUET Cyber Security Club Link
Jun 2023 to Apr 2024

Promotion and organization of cybersecurity related events, seminars and both intra and inter-university Capture The Flag (CTF) competitions multiple times throughout the tenure.

Test Scores

TOEFL: 113/120 (Reading: 29/30, Listening: 30/30, Speaking: 27/30, Writing: 27/30)
GRE: 326.0/346.0 (Verbal: 157/170, Quantitative: 165/170, AWA: 4.0/6.0)

References

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